Computer Science - Undergraduate Level Assessment 3 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Consider $x(t)$ to be given as, $ x(t)=10 $\cos(20\pi-\pi/4)-5\cos(50\pi t) Whathfsnfyetdmxvvsmrhnw$

    2. 30
    3. 20
    4. 40
    5. 50
    6. 35

Which types of functions grow the slowest?

1. O(N)
2. O(log log N)
3. O(N!)
4. O(log N)
5. O(N^N)

A 3-way, set-associative cache is

1. only effective when more than 3 processes are running alternately on the processor
2. possible only with write-through
3. one in which each main memory word can be stored at any of 4 cache locations
4. possible only with write-back
5. effective only if 3 or fewer processes are running alternately on the processor

Let x = 8. What is x>>1 in Python 3?

1. 5
2. 3
3. 0
4. 8
5. 4

In Python 3, b = [11,13,15,17,19,21]; print(b[::2]) outputs what?

13,17,21

13,15,17

11,15

11,13,15,19

11,15,19

True / False (10 marks)

Answer True or False

6. True or False: An excessively large step size can cause the training loss to increase with the num
7. True or False: In a simple planar graph with 30 vertices, the maximum number of edges is 84, ac
8. True or False: A convex polygon with 26 faces and 39 edges has 15 vertices according to Euler's
9. True or False: The TCP connect scan is designed to complete the TCP handshake for port scann
10. True or False: The use of weak classifiers in bagging helps prevent overfitting by averaging their

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to find the first repeated character in a given string.
12. Write a function to merge multiple sorted inputs into a single sorted iterator using heap queue algorithm.

End of Paper